Worksheet A: Cups, Dice and Data

Name: _____ Date: _____

Task 1: Monty Hall in pairs

in class, pairs, 10 min.

The host secretly rolls a die to place the ball (1–2: A, 3–4: B, 5–6: C), the player points at a cup, and the host lifts an *empty* cup that was not chosen. Odd rounds: stay; even rounds: switch. Tick whether the round was won.

Rd.	Strategy	Ball	won	Rd.	Strategy	Ball	won
1	stay			13	stay		
2	switch			14	switch		
3	stay			15	stay		
4	switch			16	switch		
5	stay			17	stay		
6	switch			18	switch		
7	stay			19	stay		
8	switch			20	switch		
9	stay			21	stay		
10	switch			22	switch		
11	stay			23	stay		
12	switch			24	switch		

Total: **stay** won _____ of _____ **switch** won _____ of _____ whole class (board): _____

Task 2: Means of dice: the central limit theorem

in class, individual work, 8 min.

Roll a die five times and write down the results. Compute the mean. Repeat this four times and mark all four means as dots on the number line on the board.

Series	1st	2nd	3rd	4th	5th	Mean
1						
2						
3						
4						

1. *Before* rolling: a single roll has mean 3.5 and standard deviation 1.71. What is the standard error of a mean of 5 rolls? Between which values should about 95 % of the dots on the board lie?
2. Does your prediction match the picture on the board? What shape does it have?
3. What would the picture look like if every dot were the mean of 20 rolls?

Task 3: Analysing a data set

homework, small groups

Fictitious data set (constructed for this exercise, not a real survey): 50 people held an ideomotor pendulum (cf. Unit 1). The largest swing was measured in millimetres (**mm**); afterwards the people rated how amazed they were (**W** for wonder, 1 = not at all to 5 = very much) and stated whether they had known the pendulum effect before (**PK** for prior knowledge, 1 = yes, 0 = no).

No	mm	W	PK	No	mm	W	PK	No	mm	W	PK	No	mm	W	PK	No	mm	W	PK
1	11	3	0	11	9	3	0	21	10	2	1	31	43	3	1	41	47	4	0
2	5	2	0	12	24	2	0	22	26	4	1	32	57	4	0	42	59	3	0
3	15	3	1	13	24	4	1	23	39	4	0	33	21	3	0	43	26	3	0
4	41	4	1	14	8	2	0	24	15	4	0	34	96	5	0	44	31	4	1
5	37	3	0	15	16	2	1	25	13	3	0	35	49	4	0	45	23	2	0
6	39	5	0	16	25	4	0	26	13	3	0	36	30	4	0	46	20	3	1
7	18	3	0	17	19	5	0	27	66	5	0	37	23	4	0	47	12	1	1
8	0	1	0	18	53	4	1	28	35	3	1	38	53	4	0	48	36	4	1
9	16	2	1	19	5	2	1	29	38	5	0	39	18	2	0	49	18	3	0
10	31	4	1	20	45	5	0	30	26	3	1	40	73	5	0	50	40	4	0

1. State the level of measurement of each variable.
2. For **mm**, compute the minimum, maximum, range, median, Q_1, Q_3, *IQR* and mean. State how you determined the quartiles.
3. Draw a histogram (bin width 10 mm) and a box plot. Are there outliers according to the $1.5 \cdot IQR$ rule (values more than $1.5 \cdot IQR$ below Q_1 or above Q_3)? What would you do with them?
4. Compute the mean and median of **mm** without the outlier. Which statistic reacts more strongly, and why?
5. Summarise **W** in a sensible way. Is a mean admissible?
6. Compute the standard error of the mean of **mm** and a 95 % confidence interval. What does the interval say, and what does it not say?
7. Compare **mm** between people with and without prior knowledge, using the mean *and* the median. What do you notice?
8. For the same comparison, compute a t-test by hand with the pooled standard deviation. After session 2, add the Bayes factor from the rule of thumb (Wagenmakers 2007), where N is the total number of people:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{1/n_1 + 1/n_2}}, \quad df = n_1 + n_2 - 2, \qquad BF_{01} \approx \sqrt{N} \left(1 + \frac{t^2}{df}\right)^{-N/2}$$

   What do the two results say, and what do they not say?